

Combinational Circuits

classmate

Date _____

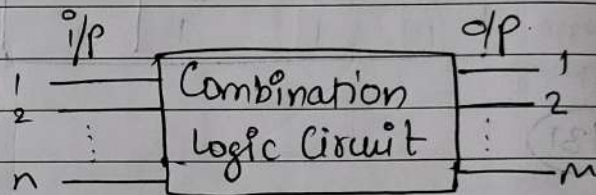
Page _____

133

- Output is dependent upon combination of input variables
- Doesn't use any memory
- It can have n number of inputs & m no. of outputs.

Example:-

- ① Adders & Subtractors
- ② Decoders
- ③ MUX
- ④ Code converters.
- ⑤ Comparator
- ⑥ ROM
- ⑦ PLA
- ⑧ PAL



134

Designing steps for a combination circuit

- ① Observe the problem definition
- ② Determine the required input & output variable
- ③ assign letters (or) symbols to the input variable
- ④ Make truth table that defines required relationship.
- ⑤ Determine the simplified boolean expression using K-map.
- ⑥ Draw logic diagram

②

classmate

Date _____
Page _____

(35) The minimal function that can detect a "divisible by 3" 8421 BCD code digit (representation is $D_8 D_4 D_2 D_1$) is given by

→

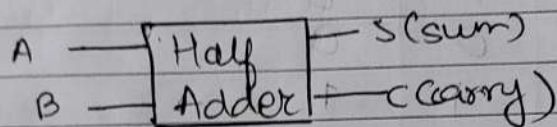
D_8	D_4	D_2	D_1	f
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1

$D_8 D_4 \backslash D_2 D_1$	00	01	10	11
00	0	1	1	2
01	4	5	7	1
11	X ₁₂	X ₁₃	X ₁₄	X ₁₅
10	8	9	X ₁₆	X ₁₇

$$f = D_8 D_1 + D_4 D_2 \bar{D}_1 + \bar{D}_4 D_2 D_1 + \bar{D}_8 \bar{D}_4 \bar{D}_2 \bar{D}_1$$

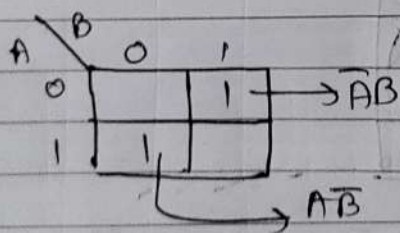
136 Designing of half Adder

- Half Adder is a combinational logic circuit designed to add two single bit no.
- It contains two inputs & two outputs (sum & carry).



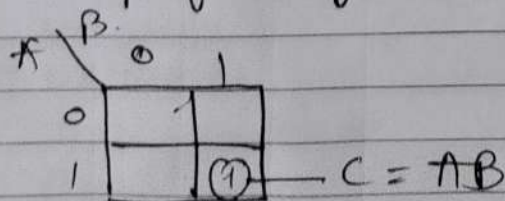
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

K-map of Sum



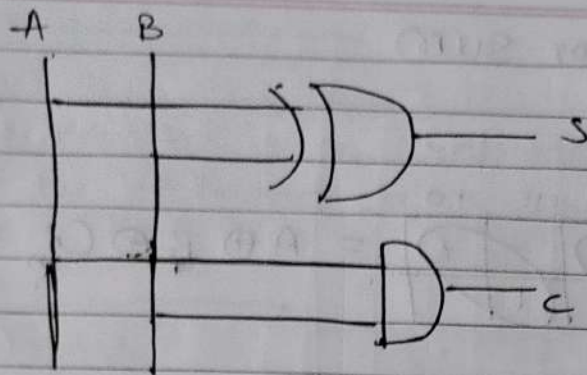
$$S = A \oplus B$$

K-map of carry.



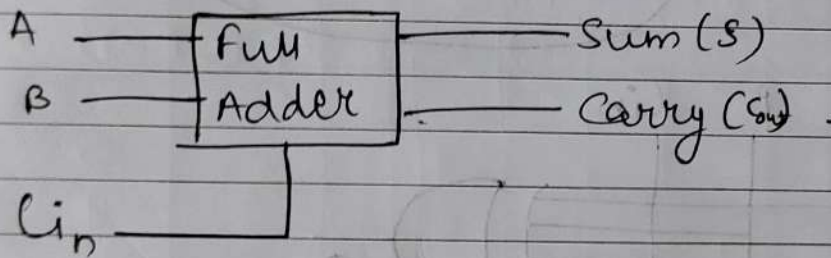
(4)

classmate

Date
Page

(137) Full Adder

Full adder is a arithmetic logic circuit designed to add two single bit no. with a carry



A	B	C _{in}	S	C _{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

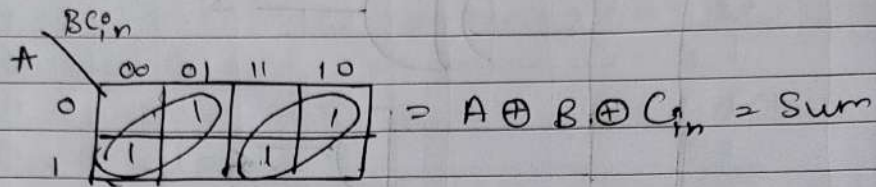
5

classmate

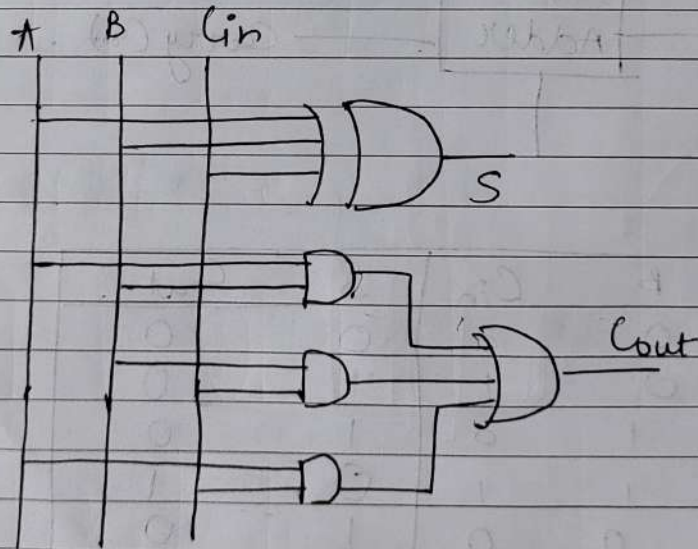
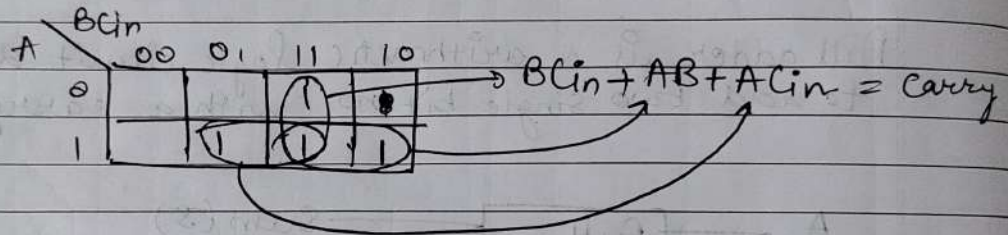
Date

Page

K-map for sum



K-map for carry



138

139

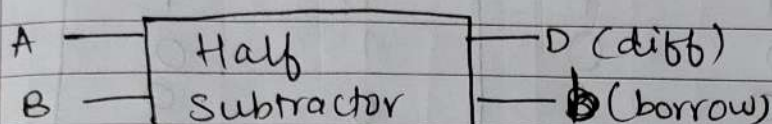
6

classmate

Date _____
Page _____

138 Half Subtractor.

Half Subtractor is a combinational circuit used to get the difference b/w two single bit no.



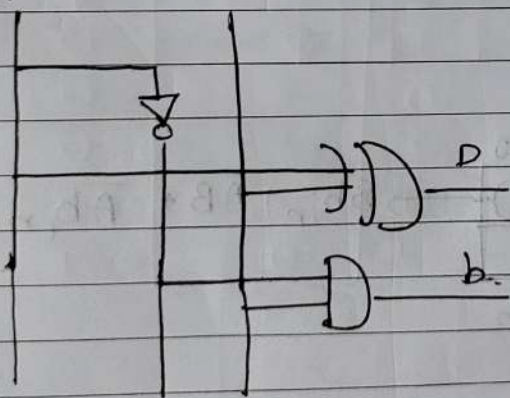
A	B	D	b
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Kmap of D.

A \ B	0	1
0	0	1
1	1	0

$\Rightarrow D = A \oplus B$

A B.



Kmap of b

A \ B	0	1
0	0	1
1	0	0

$\Rightarrow \bar{A}B$

139 Full Subtractor

Full Subtractor is combinational circuit used to perform subtraction among 3 bits.

7

classmate

Date _____
Page _____

A	B	bin	D	bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

K map of D.

B bin

A	00	01	11	10
0	0	1	1	1
1	1	1	1	1

$$\Rightarrow D = A \oplus B \oplus \text{bin}$$

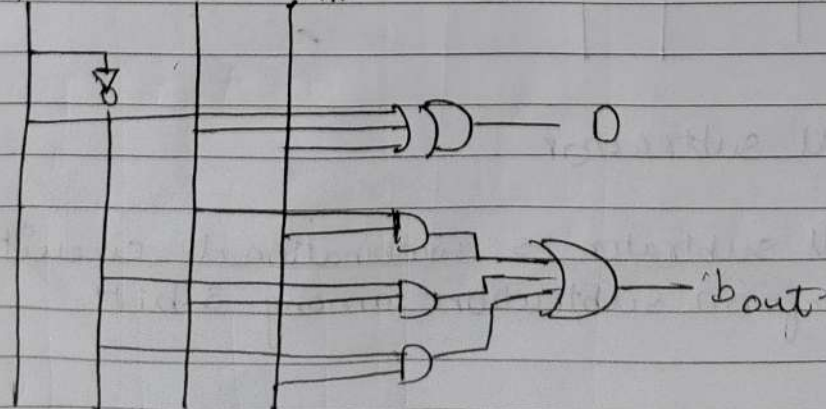
K map of bout

B bin

A	00	01	11	10
0	0	1	1	1
1	1	1	1	1

$$\Rightarrow \text{bout} = B \text{bin} + \bar{A}B + \bar{A}\text{bin}$$

A B bin



⑧

Also called
Ripple carry adder

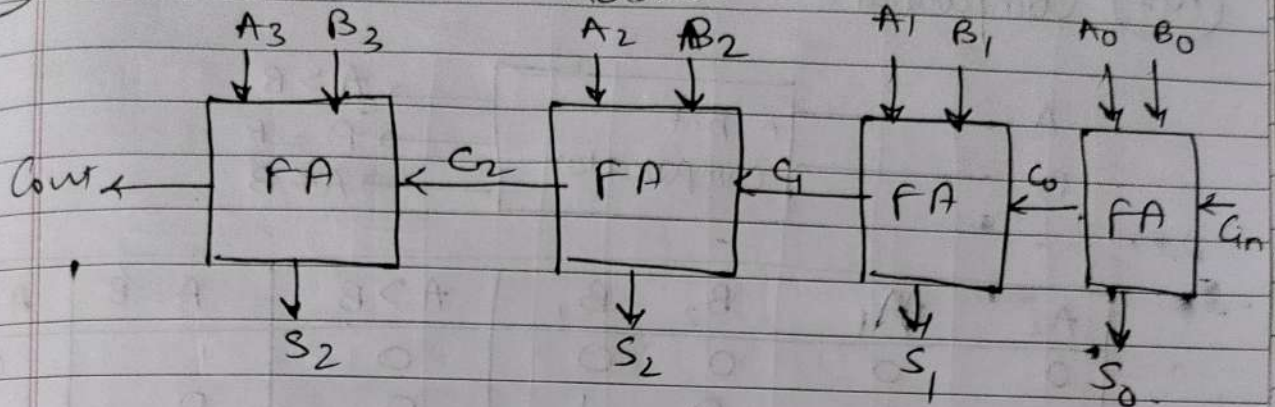
classmate

Date

Page

140

N-Bit Parallel Adder



For full adder, $\text{Sum} = A \oplus B \oplus C$
 $\text{Carry} = AB + BC + CA$

141

4-bit Parallel Adder cum Subtractor

142

Full Adder using half adder.

143

Carry generation in carry look ahead adder

144

Carry look Ahead Adder.

145

BCD Adder.

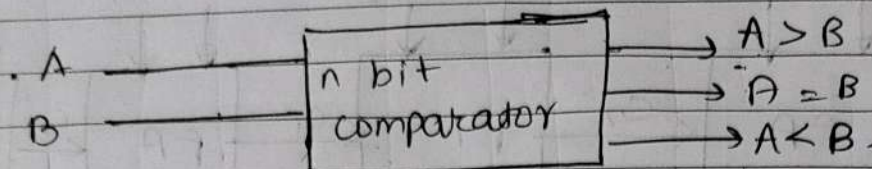
146

Excess-3-Adder

147

Comparator

(147) Comparator.



A_2	A_1	B_2	B_1	$A > B$	$A = B$	$A < B$
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	0	0	0
1	0	1	0	0	0	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

(148) K-map for $A > B$

$B_2 B_1$ \ $A_2 A_1$	00	01	11	10
00		1	1	1
01			1	1
11				
10			1	

$$\Rightarrow \overline{B_2} A_2 + B_1 A_2 A_1 + \overline{B_2} \overline{B_1} A_1$$

(10)

classmate

Date

Page

K map for $A = B$.

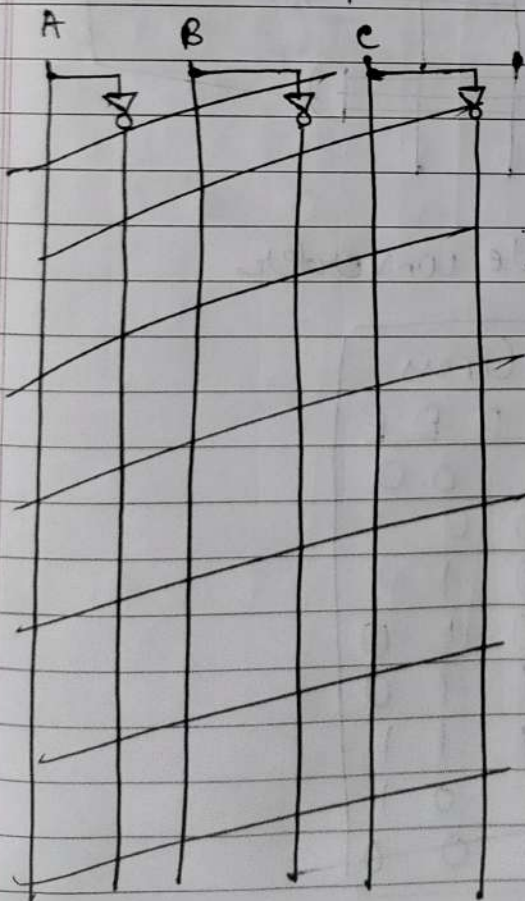
$B_2 B_1 \backslash A_2 A_1$	00	01	11	10
00	1			
01		1		
11			1	
10				1

$$\Rightarrow (A_2 \odot B_2)(A_1 \odot B_1)$$

K map for $A < B$

$B_2 B_1 \backslash A_2 A_1$	00	01	11	10
00				
01	1			
11	1	1		1
10	1	1		

$$\Rightarrow B_2 \bar{A}_2 + B_1 \bar{A}_2 \bar{A}_1 + B_2 B_1 \bar{A}_1$$



000

001

010

011

100

101

110

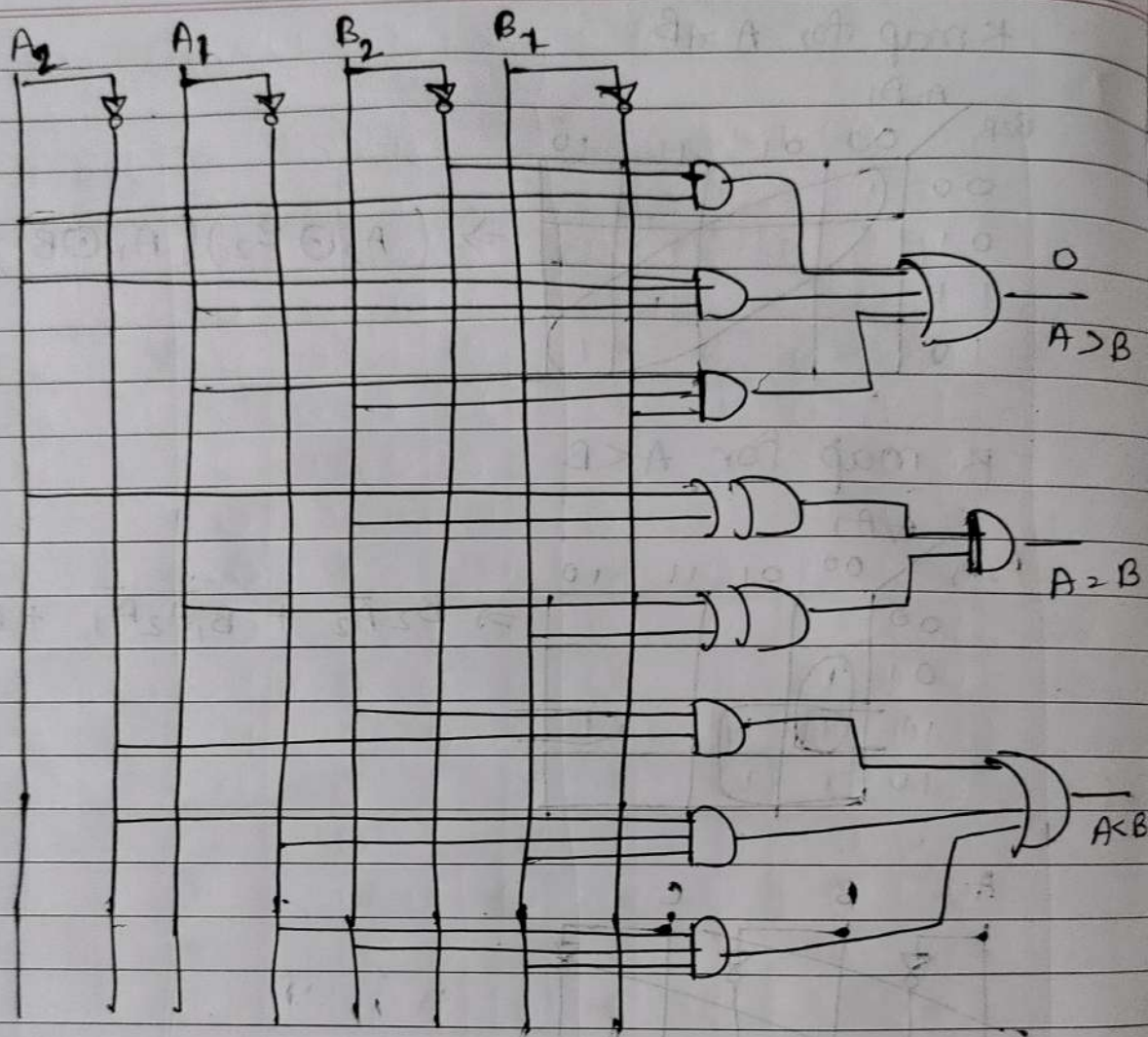
111

(11)

classmate

Date _____

Page _____



(149)

Binary to Gray code converter

Decimal	Binary A B C	Gray D E F
0	0 0 0	0 0 0
1	0 0 1	0 0 1
2	0 1 0	0 1 1
3	0 1 1	0 1 0
4	1 0 0	1 1 0
5	1 0 1	1 1 1
6	1 1 0	1 0 1
7	1 1 1	1 0 0

(12)

classmate

Date

Page

K map for D.

A \ BC				
	00	01	11	10
0				
1	1	1	1	1

$$\Rightarrow D = A$$

K map for E.

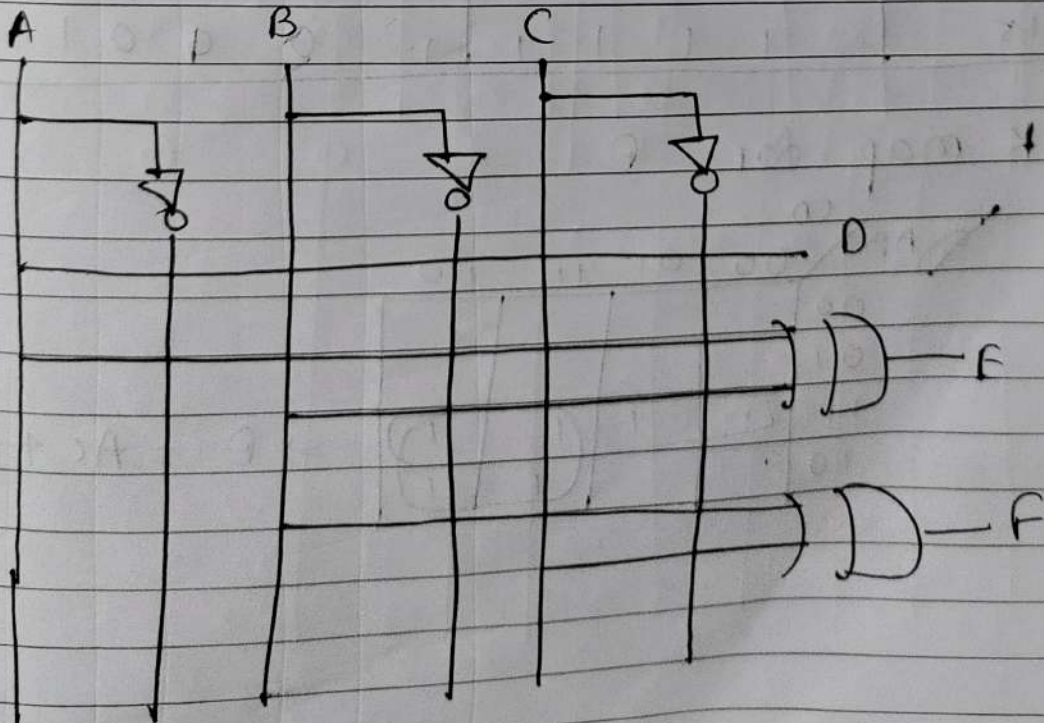
A \ BC				
	00	01	11	10
0			1	1
1	1	1		

$$E = \bar{A}B + A\bar{B} = A \oplus B$$

K map for F.

A \ BC				
	00	01	11	10
0		1		1
1		1		1

$$F = \bar{B}C + B\bar{C} = B \oplus C$$



13

classmate

Date

Page

150 Code converter binary to BCD.

Decimal	Binary				Binary BCD				
	A	B	C	D	E	F	G	H	I
0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	1
2	0	0	1	0	0	0	0	1	0
3	0	0	1	1	0	0	0	1	1
4	0	1	0	0	0	0	1	0	0
5	0	1	0	1	0	0	1	0	1
6	0	1	1	0	0	0	1	1	0
7	0	1	1	1	0	0	1	1	1
8	1	0	0	0	0	1	0	0	0
9	1	0	0	1	0	1	0	0	1
10	1	0	1	0	1	0	0	0	0
11	1	0	1	1	1	0	0	0	1
12	1	1	0	0	1	0	0	1	0
13	1	1	0	1	1	0	0	1	1
14	1	1	1	0	1	0	1	0	0
15	1	1	1	1	1	0	1	0	1

K map for E

AB	CD			
	00	01	11	10
00				
01				
11	1	1	1	1
10			1	1

$$\Rightarrow E = AC + AB$$

W

* map for F.

AB \ CD	00	01	11	10
00				
01				
11				
10	1	1		

$$F = A \bar{B} \bar{C}$$

K map for G

AB \ CD	00	01	11	10
00				
01	1	1	1	1
11			1	1
10				

$$G = BC + \bar{A}B$$

K map for H

AB \ CD	00	01	11	10
00			1	1
01			1	1
11	1	1		
10				

$$H = \bar{A}C + AB\bar{C}$$

K map for I

AB \ CD	00	01	11	10
00		1	1	
01		1	1	
11		1	1	
10		1	1	

$$I = D$$

for diagram
see vid. 180.

(15)

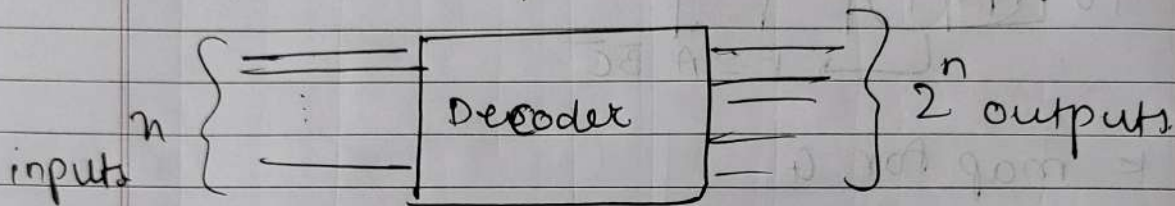
classmate

Date

Page

(154) Decoder (2 to 4 Decoder)

Decoder is a multi input - multioutput logic circuit which decodes n inputs into 2^n possible outputs.



where $E = \text{Enable}$

- $\nearrow E=0$ - Decoder disabled
- $\searrow E=1$ - Decoder enabled

			2^3	2^2	2^1	2^0
E	A	B	Y_3	Y_2	Y_1	Y_0
0	X	X	0	0	0	0
1	0	0	0	0	0	1
	0	1	0	0	1	0
	1	0	0	1	0	0
	1	1	1	0	0	0

$$Y_3 = AB$$

$$Y_2 = A\bar{B}$$

$$Y_1 = \bar{A}B$$

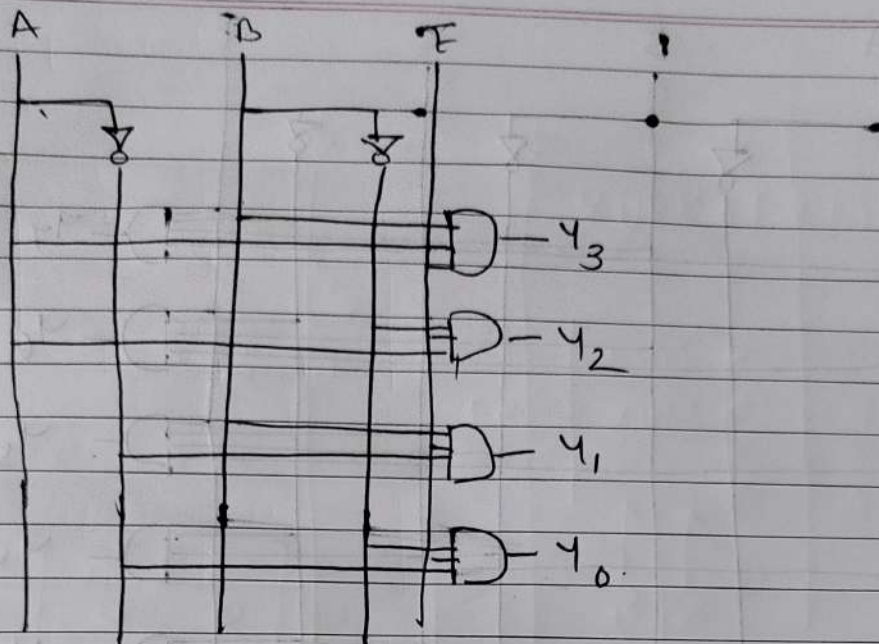
$$Y_0 = \bar{A}\bar{B}$$

(16)

classmate

Date

Page

(155) Decoder (3 to 8)

$$n=3; 2^n = 2^3 = 8$$

				2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
E	A	B	C	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	x	x	x	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	1
1	0	0	1	0	0	0	0	0	0	1	0
1	0	1	0	0	0	0	0	0	1	0	0
1	0	1	1	0	0	0	0	1	0	0	0
1	1	0	0	0	0	0	1	0	0	0	0
1	1	0	1	0	0	1	0	0	0	0	0
1	1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	1	0	0	0	0	0	0	0

$$Y_7 = ABC$$

$$Y_2 = \overline{A}BC$$

$$Y_6 = AB\overline{C}$$

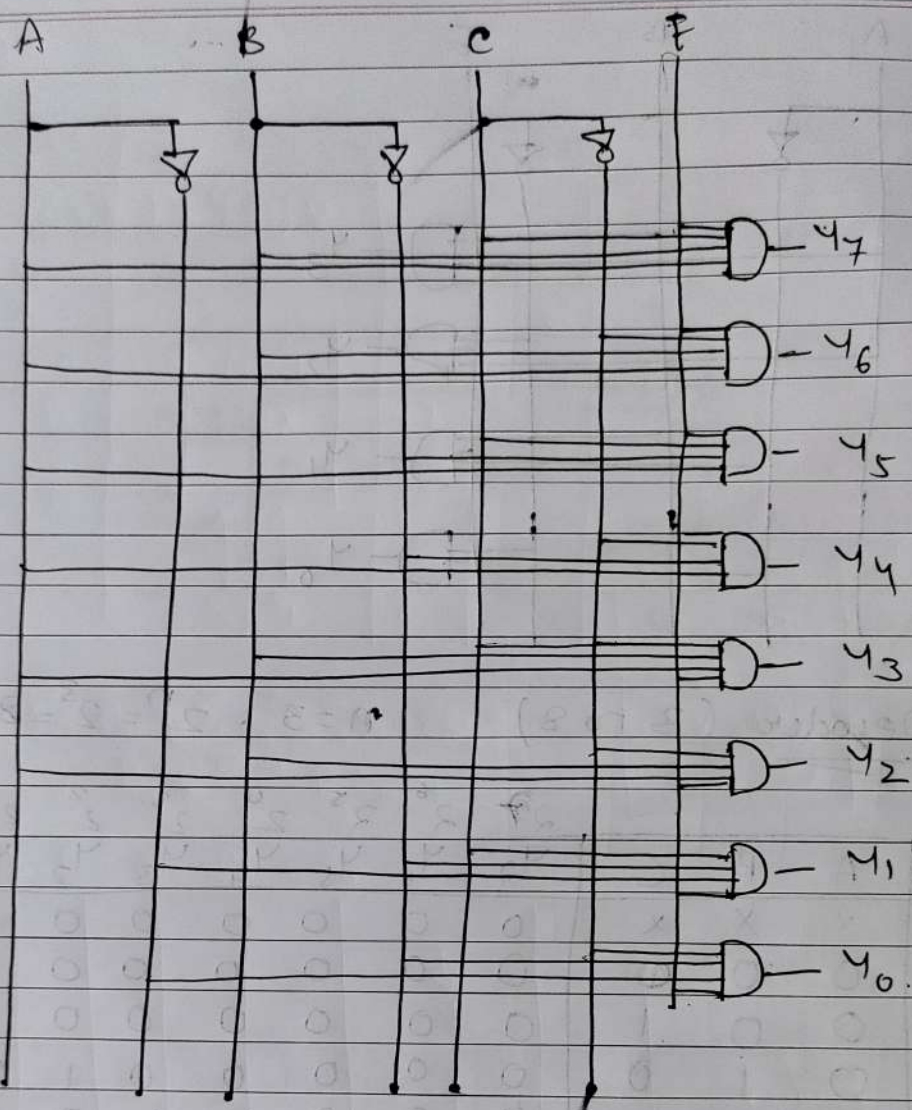
$$Y_1 = A\overline{B}C$$

$$Y_5 = A\overline{B}\overline{C}$$

$$Y_0 = \overline{A}\overline{B}\overline{C}$$

$$Y_4 = A\overline{B}C$$

$$Y_3 = \overline{A}BC$$



156 Multiplexer (MUX).

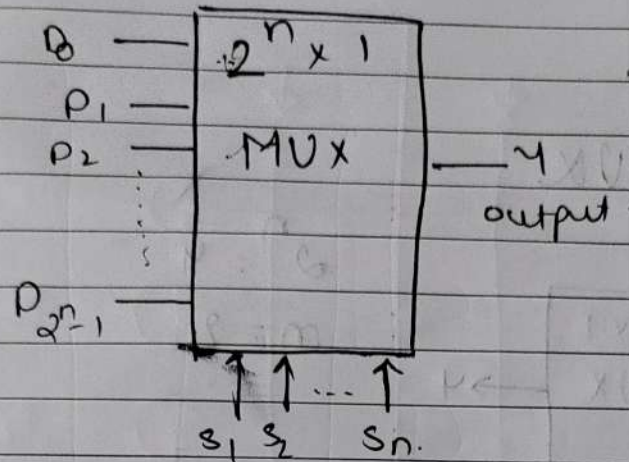
- Multiplexer is a combinational logic circuit used to select only one input among several inputs based on selection line.
- This can act as digital switch.
- This is also called as data selector.
- For Mux there can be 2^n inputs, n selection line, and only one output.

(18)

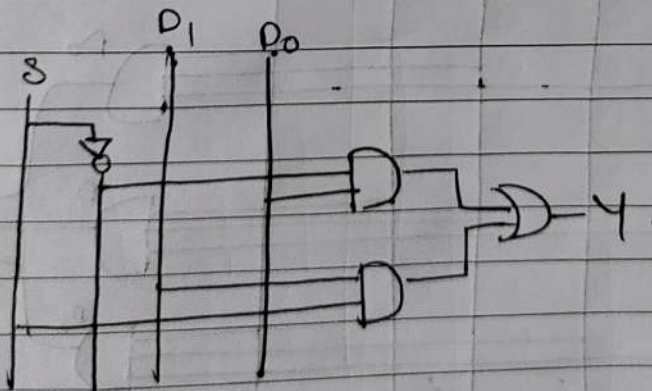
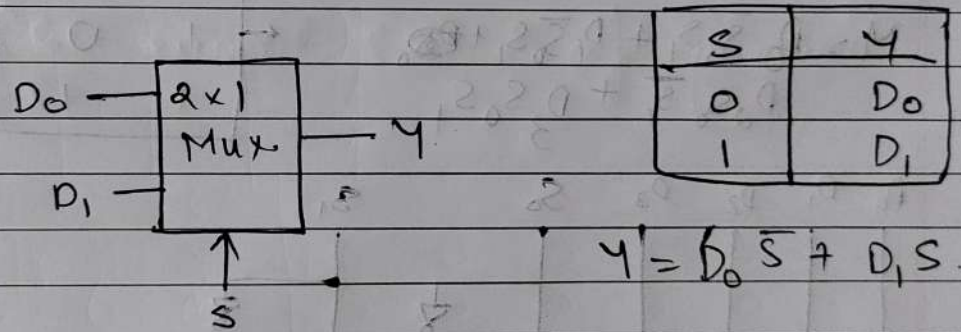
classmate

Date

Page



Example 2×1 Mux. Design.



(157)

$A \odot B \rightarrow \text{Ex-NOR}$

$A \oplus B \rightarrow \text{Ex-OR}$

$000 = 1$
 $001 = 0$
 $000 = 0$
 $001 = 1$

19

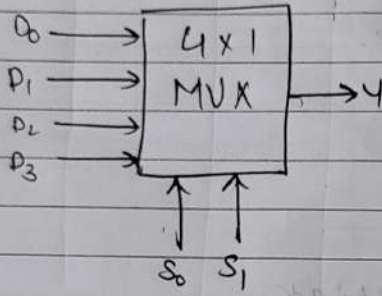
classmate

Date _____
Page _____

158

159

4x1 MUX

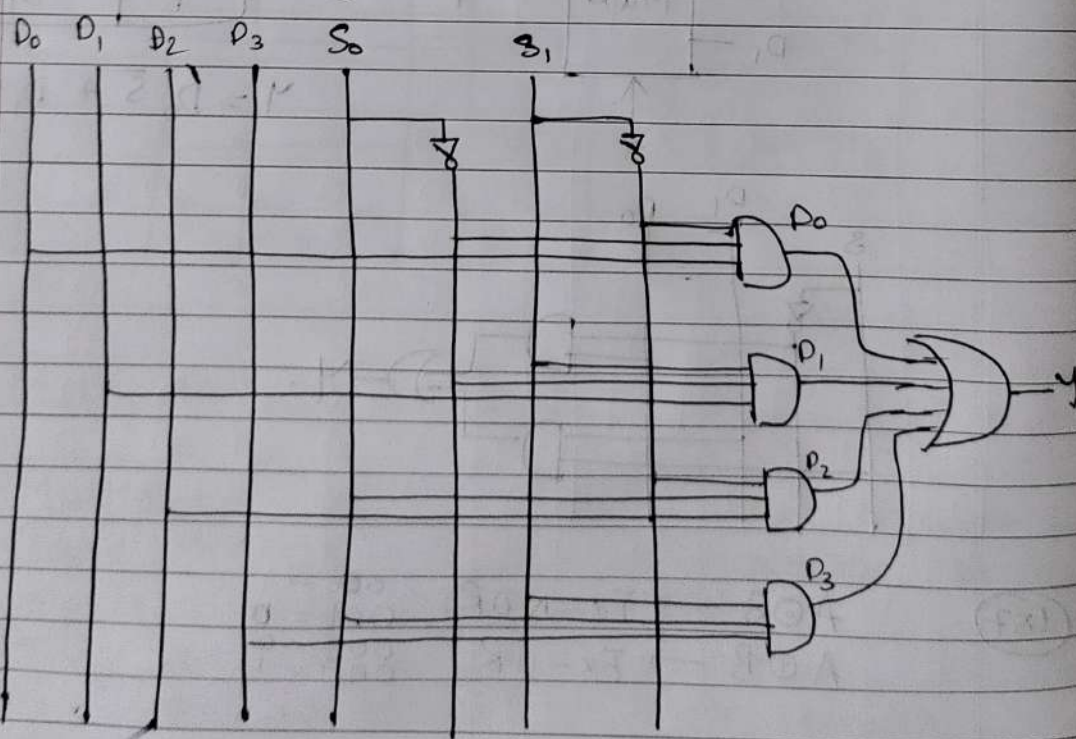


$$2^n = 4$$

$$n = 2$$

S_0	S_1	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

$$Y = D_0 \bar{S}_0 \bar{S}_1 + D_1 \bar{S}_0 S_1 + D_2 S_0 \bar{S}_1 + D_3 S_0 S_1$$



160

161

162

Not in Videos

20

15

classmate

Date

Page

160

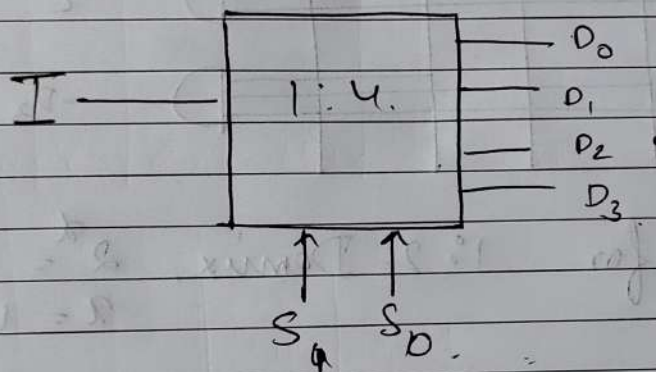
161

162

Not in
Videos

Demux $\Rightarrow$ 1:n.

1:4 Demux. $2^x = 4 \therefore x = 2$



S ₁	S ₀	D ₀	D ₁	D ₂	D ₃
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

$$D_0 = I \cdot \overline{S_1} \cdot \overline{S_0}$$

$$D_1 = I \cdot \overline{S_1} \cdot S_0$$

$$D_2 = I \cdot S_1 \cdot \overline{S_0}$$

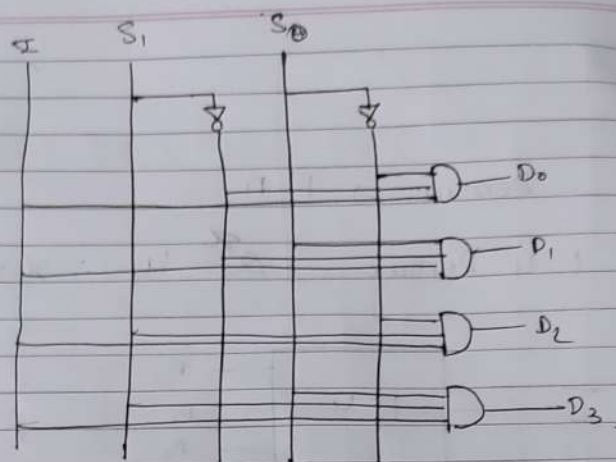
$$D_3 = I \cdot S_1 \cdot S_0$$

(21)

classmate

Date

Page



Similarly for 1:2 Demux. $2^x = 2$
 $2 = 1$

Encoder

 $2^n \rightarrow n$

i/p o/p

8 to 3 encoder

D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	x	y	z
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

22

classmate

Date

Page

$$x = D_4 + D_5 + D_6 + D_7$$

$$y = D_2 + D_3 + D_6 + D_7$$

$$z = D_1 + D_3 + D_5 + D_7$$

